Assignment 7.1: Final Digital Storytelling Case Study

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ESH 531: Environmental Justice

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August 10, 2026

Alviso is a small town located in North San Jose, near the water South of San Francisco Bay. The area is home to a historic port town sandwiched between the Silicon Valley and the salt flats along with other nature. Alviso was founded in the nineteenth century as a transportation and agricultural center. It also became the Port of San Jose as it connected Santa Clara Valley to San Francisco Bay. The community joined the area with a diverse community in 1852 until it was annexed in 1968 (Jose, 2024). Along with the diverse community, Alviso also had a diverse history of industries including cannery, steamboat, railroad, agriculture, salt production, industrial development, and many flooding events.

The environmental vulnerability that Alviso faces is not just because of its geography or climate change. It is a result of the historic development, environmental degradation, unequal politics, and decisions making that has shifted environmental harm on the marginalized community in Alviso. Applying environmental justice, risk society, and Verchick's feminist method theories show that making Alviso more resilient needs technical investment in flood protection and wetland restoration as well as community participation and acknowledging the historic inequities that made the community vulnerable.

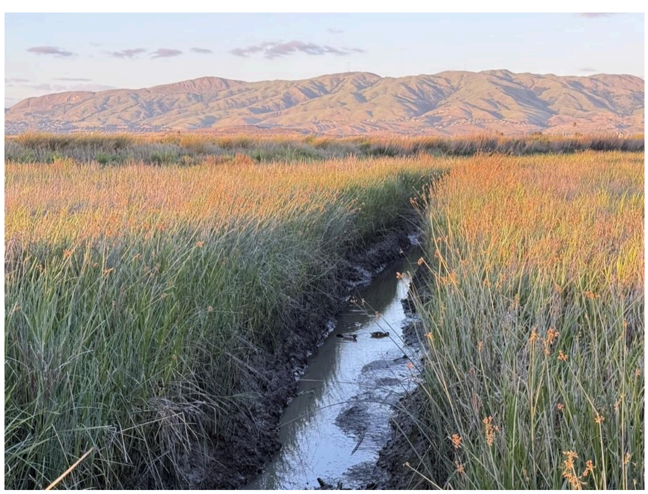

Image of Coyote Creek taken by Snigdha Gurram.

Historical Context

To understand the environmental conditions of present Alviso there needs to be an understanding of how it started in the beginning. Alviso was first home to the Muwekma Ohlone Tribe and Tamien people who were located throughout what is now known as the Bay Area (ThemeZaa, n.d.). Later Spanish missionaries took over this land and it was slowly transformed over the years into Mission Santa Clara from 1777 to 1840 (Jose, 2024). There was a diverse community that grew with Chinese, Japanese, Portuguese, and Mexicans who lived there for various reasons and industries throughout the years (ibid.). Alviso went through many different businesses in the past, these include cannery, steamboat, railroad, and agriculture (ibid.).

Image of a drawing of the Muwekma Ohlone Tribe. From https://storymaps.arcgis.com/stories/f68a710a71c048189d9b209a4d966338

This history is important because it shapes a relationship between the environmental changes and the economic developments. One of the more transformative occurrences was through the salt production. USGS research shows that by 1930 about half of the historical tidal marshes were changed into salt ponds in the South Bay (South Bay Salt Pond Restoration Project, n.d.). The ponds in Alviso became a part of an industry that changed the hydrology and landscape of the South San Francisco Bay (Masters degree candidates of the Urban & Regional Planning Department, 2009).

Land subsidence also added to Alviso vulnerability. Groundwater was being pumped to use the water for agricultural purposes and urban development (Masters degree candidates of the Urban & Regional Planning Department, 2009). As the water was pumped over the years, the land started to sink to about six feet between the late 1800s and 1970 (ibid.). Currently some parts of Alviso are around 13 feet below sea level (Jose, 2024). The California Department of Water Resources talks about Santa Clara Valley as the best example of economic and environmental ground water related consequences in California (Santa Clara Valley Water, 2019). The geography of the valley being sloped towards the San Francisco Bay makes the subsidence increase the consequences of coastal flooding. The Alviso area faced serious flooding in 1983, which shows that flood control infrastructures could be inadequate as the conditions change.

Steamboat accident memorial plaque in Alviso. Picture by Snigdha Gurram.

Theoretical Framework

Environmental Justice

The first theoretical framework for Alviso is environmental justice. Robert Bullard challenged the thought that environmental protection can be separated from race, class, political power, and human rights (Bullard, 2005). Environmental justice is about who receives benefits, experiences burdens, is part of the decision making process, and is recognized by institutions (ibid.). This framework can be split into three areas: distributive justice, procedural justice, and recognition justice.

Distributive justice is about environmental problems and benefits being shared equally. In Alviso, the environmental burdens are flooding, vulnerable to sea level rise, land subsidence, wetland changes, and distance to infrastructures. Alviso is in an economically filled area that benefits from transportation networks, industrial development, technology, and growth of the whole area in general. Distributive justice motivates people to focus on how the cost and benefits of development are spread out.

Procedural justice is about whether people have good opportunities to help make decisions that affect their lives. This is important for a place like Alviso because decisions affecting their future have many officials involved whose voices are more prevalent. Flood protection, restoration of wetlands, transportation, land use, wastewater management, and climate adaptation can include the City of San Jose, Santa Clara Valley Water, state agencies, federal agencies, and other organizations (City of San Jose, n.d.). This shows that the historical documentation can be an environmental justice practice since decisions made without understanding the history of what people face, then the outcomes would still likely cause more inequalities that strengthen the vulnerability of the community.

Justice as recognition is about showing respect to identities, cultures, history, and experiences of communities that are most affected by the harm. Alviso is not just a geographical area that is below sea level or an engineering problem. It is a strong historic port town that has withstood many of the damaging impacts that have come its way. So this means recognizing the value of the community itself.

Risk Society

The next theoretical framework for Alviso is risk society. A risk society is about unintended consequences of industrialization and technology development causing environmental problems and health problems that are not equally seen (Cable et al., 2009). For Alviso, climate and environmental problems are because of natural events caused by decisions made by humans about the use of the land, infrastructure, and allocation of resources. The community members are told to be prepared for the floods, improve housing or medical assistance, without resources from the government. The residents have to do this while they did not cause the problem nor having their other problem solved.

Alviso land subsidence in the 20th century and 1970s. From https://www.valleywater.org/your-water/groundwater/subsidence

Verchick's Feminist Method

The final theoretical framework for Alviso is Verchick's Feminist Method. Verchick talks about how the feminist method contains important frameworks to understand environmental inequality through unmasking, contextual reasoning, and consciousness raising (Verchick, 1996).

Unmasking tells us to question policies and decisions around the environment that are neutral but actually back up inequalities (Verchick, 1996). Alviso has faced many environmental problems since it is near various industries (Masters degree candidates of the Urban & Regional Planning Department, 2009). The decisions made about the wetland changes, urban development, and flood control are thought to be necessary for economic growth but this instead made Alviso more vulnerable (ibid.).

Contextual reasoning tells us to understand environmental problems through real experiences from the most impacted areas (Verchick, 1996). Alviso is a historical town that has deep rooted connections. The experience from the community have documented their concerns over the years (Masters degree candidates of the Urban & Regional Planning Department, 2009).

Some of the concerns were about flood events, improper infrastructure, and sea level rise affecting the community (ibid.). These experiences have important perspectives that can not be found in science.

Consciousness raising tells us that personal experiences that are similar can help build awareness and inspiration towards action (Verchick, 1996). There have been organizations, public meetings, and local advocacy showing the communities efforts towards bringing awareness to flooding, wetland restoration, and climate adaptation. These public meetings let the community members talk about their experiences with people in the same situation and with more environmental justice issues.

This is an image of a community meeting in Alviso in 2026. From https://www.kneedeeptimes.org/united-alviso-for-change-climate-resilience/

Analysis

These three theories are not separate problems from each other, but instead three different parts of the same root problem. Flooding is an infrastructure and public health problem that include possible problems like injury, stress, displacement, mold, contaminated water, disruption over various services. The South San Francisco Bay Shoreline Project sees that there needs to be protection for the community and the infrastructure (Santa Clara Valley Water, n.d.). Flooding can mix with contaminants which has been studied to show an increase in environmental injustice from climate related flood events for vulnerable communities that are being exposed (Liu & Mostafavi, 2023). Adaption should have water quality and hazardous material monitoring plans alongside levee building.

Public health is involved in mental, physical, social well being. Stress can be caused from warnings, uncertainty, and fear. Residents do not have access to resources to recover on their own without the help of outside resources. A strategy that is community centered should be multilingual, accessible, and support households that can not afford it.

The analysis works towards resilient strategies that are structural and participatory. To start, flood protection should continue as the South San Francisco Bay Shoreline Project is one of the biggest investors in it with 1-3 being completed in 2025 (Santa Clara Valley Water, n.d.). The second strategy can be for the residents to have a lot more power in decision making and adaptation planning. A Alviso Climate Resilience Advisory Group can help bring residents, planners, officials, Valley Water, environmental organizations, and businesses together to go over flood data and communicate with each other on steps that should be taken from each group. The third strategy can be for information to be publicly available. Agencies and other officials should provide information in simple language so that everyone can understand. From the flood maps to limitations, they should be accessible, multilingual, and simple for the residents to understand. The fourth strategy can be for the investment to clearly address what they are funding. Funding should support the community in all aspects. If flood protection increases the value of the property, the possibility of displacement should be monitored carefully and strategies to prevent it should be considered. Lastly, Alviso should be treated as a community just like all the other communities in the Silicon Valley instead of an area that needs help to prevent flooding. The City of San Jose project shows that the residents have knowledge that is not stated in the planning documents (City of San Jose, 2026). The three theories combined show that resilience should protect the people, ecosystem, and community as a whole in every aspect.

South San Francisco Bay Salt Pond Project Before and After Image. From https://www.southbayrestoration.org/

Conclusion

Alviso shows how environmental vulnerability is not as simple as one might think. Alviso is located near the San Francisco Bay which exposes it but because of the development decisions in the past made it more vulnerable and made the exposure more severe. Development of the port, agriculture, extraction, transportation, industries, and alteration for economic benefit changed the whole landscape. The environmental justice from Bullard explains that race, class, political power, and human rights are all related to the inequality the community faces. Risk society explains why hazards are complicated and why the responsibility is moved to the people already facing the problem. Verchick's feminist method explains why technical systems can still cause injustice when experiences are not taken into consideration. These theories support the thesis that Alviso being vulnerable to flooding is socially made environmental injustice because of development, resource use, and informational power that is withheld. The shoreline project shows that a solution is possible with engineered protection and restoration of ecology together. Community engagement and historic documentation shows that institutions can further their knowledge for planning. For my MESH project, this case has changed the way I think about engineering and sustainability. Technical feasibility is important but not enough while sustainable solutions have to include history, power, health, and equity. Instead of focusing on how high the levee should be for example, planners should think about who is protected, participating, what information is available, and what happens to the residents and environment. The future of Alviso is not just protecting the community from the flooding but building a community with the power, resources, and information to adapt to changes.

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